

ACADEMIC AND PROFESSIONAL EXPERIENCE

New Jersey Institute of Technology	
Tenure-track Assistant Professor	2025–current
Technion - Israel Institute of Technology	
Senior Lecturer (tenure-track Assistant Professor)	2023–2024
Columbia University	
Term Assistant Professor in Applied Mathematics and Associate Research Scientist	2019–2023
Yale University	
Visiting Graduate Student	2018–2019
Tel Aviv University	
Junior Lecturer and Teaching Assistant	2016–2018

EDUCATION

Tel Aviv University	
Ph.D. in Applied Mathematics	2016–2019
Tel Aviv University	
M.Sc. in Applied Mathematics	2014–2016
Hebrew University of Jerusalem	
B.Sc. in Mathematics and Physics (“Talpiot” program)	2006–2009

GRANTS AND AWARDS

- NSF DMS-2508811 (\$182,956) “A Dynamical Theory of Floquet Materials,” PI, 2025–2027.
- Binational Science Foundation (BSF) Grant No. 2022254 (\$118,000) “Floquet Media - a Dynamic and Spectral Approach,” PI, with M.I. Weinstein 2023–2027*
- * since only Israeli PI’s are eligible, returned in 10/24 upon moving to NJIT.
- AMS-Simons Travel Grant (\$5,000) 07/21–07/23
- SIAM Early Career Travel Award (CSE25) 03/25
- SIAM Early Career Travel Award (CSE21) 03/21
- SIAM Student Travel Award (CSE19) 02/19
- Israel Ministry of Science and Technology Doctoral Student 11/18
- Tel Aviv University Distinguished Ph.D. Award (School of Mathematics) 06/18
- SIAM Student Travel Award (NWCS18) 06/18
- Tel Aviv University Distinguished M.Sc. Award (School of Mathematics) 05/15
- Dean’s List Excellence Award (Hebrew University of Jerusalem) 03/09

1. M.B. Erdogan, J. Kraisler, and **A. Sagiv**, “Dispersive Estimates for Dirac Operators with General Domain Walls: One and Two Dimensions.” *arXiv:2607.22879*.
2. R. Kassem, **A. Sagiv**, and M.I. Weinstein, “Dispersive decay bounds for the SSH model on the half-line.” *arXiv:2605.08372*.
3. A. Bloch, **A. Sagiv**, and S. Steinerberger, “Slow Dispersion in Floquet-Dirac Hamiltonians.” *arXiv:2603.28715*. Accepted for publication at **SIAM J. on Applied Mathematics**.
4. A. Chertock, P. Degond, **A. Sagiv**, and L. Wang, “The Evolution of Pointwise Statistics in Hyperbolic Equations with Random Data.” *arXiv:2507.11399*. **SIAM/ASA J. on Uncertainty Quantification**, 14.3, 1045–1079, 2026.
5. **A. Sagiv**, R. Kassem, and M.I. Weinstein, “Dispersive Decay Estimates for periodic Jacobi operators on the half-line.” **J. of Mathematical Analysis and Applications**, 553, 129945, 2026.
6. J. Kraisler, **A. Sagiv**, and M.I. Weinstein, “On the Time-decay of solutions arising from periodically forced Dirac Hamiltonians.” **J. of Differential Equations**, 440, 113449, 2025.
7. **A. Sagiv** and M.I. Weinstein, “Near invariance of quasi-energy spectrum of Floquet Hamiltonians.” **J. of Spectral Theory**, 16, 425–458, 2026.
8. R. Baptista, B. Hosseini, N.B. Kovachki, Y.M. Marzouk, and **A. Sagiv**, “An Approximation Theory Framework for Measure-Transport Sampling Algorithms.” **Mathematics of Computation**, 94, 1863–1909, 2025.
9. J. Kraisler, **A. Sagiv**, and M.I. Weinstein, “Dispersive decay estimates for Dirac equations with a domain wall.” **SIAM J. on Mathematical Analysis**, 56, 7194–7227, 2024.
10. Q. Du and **A. Sagiv**, “Minimizing optimal transport for functions with fixed-size nodal sets.” **J. of Nonlinear Science**, 33:95, 2023.
11. S.N. Hameedi, **A. Sagiv**, and M.I. Weinstein, “Radiative decay of edge states in Floquet media.” **SIAM Multiscale Modeling and Simulations**, 21, 925–962, 2023.
12. **A. Sagiv**, “Spectral Convergence of Probability Densities for Forward Problems in Uncertainty Quantification.” **Numerische Mathematik** 150, 1165–1185. 2022.
13. **A. Sagiv** and M.I. Weinstein, “Effective Gaps in Continuous Floquet Hamiltonians.” **SIAM J. on Mathematical Analysis**, 54, 986–1021, 2022.
14. O. Lindenbaum, **A. Sagiv**, G. Mishne, and R. Talmon, “Kernel-Based Parameter Estimation of Dynamical Systems with Unknown Observation Functions.” **Chaos: An Interdisciplinary Journal of Nonlinear Science**, 31, 043118, 2021.

15. **A. Sagiv** and S. Steinerberger. “[Transport and Interface: an Uncertainty Principle for the Wasserstein Distance.](#)” **SIAM J. on Mathematical Analysis**, 52, 3039–3051, 2020.
16. **A. Sagiv**, A. Ditkowski, R.H. Goodman, and G. Fibich. “[Loss of Physical Reversibility in Reversible Systems.](#)” **Physica D**, 410, 132515, 2020.
17. **A. Sagiv**. “[The Wasserstein Distances Between Pushed-Forward Measures with Applications to Uncertainty Quantification.](#)” **Communications in Mathematical Sciences**, 18, 707–724, 2020.
18. A. Ditkowski, G. Fibich, and **A. Sagiv**. “[Density Estimation in Uncertainty Propagation Problems Using a Surrogate Model.](#)” **SIAM/ASA J. on Uncertainty Quantification**, 8, 261–300, 2020.
19. G. Patwardhan, X. Gao, **A. Sagiv**, A. Dutt, J. Ginsberg, A. Ditkowski, G. Fibich, and A.L. Gaeta. “[Loss of Polarization of Elliptically Polarized Collapsing Beams.](#)” **Physical Review A**, 99, 033824, 2019.
20. **A. Sagiv**, A. Ditkowski, and G. Fibich. “[Loss of Phase and Universality of Stochastic Interactions Between Laser Beams.](#)” **Optics Express**, 25, 24387–24399, 2017.

TALKS

Invited and Seminar Talks

- 2027: Workshop on **Mathematical Analysis and Modeling of Maxwell’s Equations in Complex Media**, at ICERM.
- 2026: **JMM ILAS** special session on Model Order Reduction and Data-Driven Modeling (Washington, DC), **Indiana University** (applied math seminar), **15th Oxbridge PDE Conference** (Oxford University), **SIAM UQ26** (invited minisymposium talk on measure transport for inverse problems and ML) at Minneapolis, MN, **Baylor Analysis Seminar**, **Drexel applied math seminar**, **CUNY Graduate Center** Harmonic Analysis and PDE seminar, **9th International Conference on Computational Harmonic Analysis** at Vanderbilt University, **IMSI workshop** at the University of Chicago “Mathematical Aspects of 2D Quantum Materials and Meta-materials”, **Bielefeld University** Analysis Seminar (via zoom), **Brin Center workshop** “Structure-Preserving Numerical Methods for Nonlinear PDEs with Uncertainty” at the University of Maryland. **SIAM MDS** (invited minisymposium talk), Salt Lake City, UT.
- 2025: **UIUC** (PDE seminar), **Rutgers** (applied math), **ADMOS25** (minisymposium on adaptive sampling and surrogate modelling, Barcelona), **SIAM AN25** (minisymposium on analysis and modeling in photonics, Montreal), **Rice CMOR Colloquium**, **Auburn** (applied and comp math seminar), **SIAM PD25** (minisymposium talk on novel materials + minisymposium talk on Statistical Approaches to PDE Inverse Problems, Pittsburgh), **Workshop on Mathematical Analysis in Machine Learning** at Pittsburgh University
- 2024: **Early Career Workshop in Mathematical Physics** (at Texas A&M), **AMS Fall Central Sectional** (at UTSA), **Hebrew University of Jerusalem** (analysis), **Tel Aviv University** (applied math), **U of Washington** (applied math), **Texas A&M** (colloquium), **NJIT** (colloquium), **UBC** (colloquium)

- **2023:** **Louisiana State** (math physics, online), **Stony Brook** (analysis), **SIAM NNP Sectional** (at NJIT), **Michigan** (PDE), **Michigan State** (math physics), **UC San Diego** (math of data science), **UC Santa Barbara** (applied math), **Yale** (analysis), **Princeton** (PACM colloquium)
- **2022:** **Maryland** (numerical analysis), **CUNY** (harmonic analysis & PDE), **Minnesota** (applied math), **South Carolina** (applied math, online), **NJIT** (waves seminar), **SIAM Annual Meeting** (at Pittsburgh), **TU Chemnitz** (stochastics seminar, online), **UC Davis** (Center of Quantum Mathematics and Physics, math seminar), **Workshop talk:** Approximation of high-dimensional parametric PDEs in forward UQ workshop (Erwin Schrödinger Institute, online), **U of Washington** (applied math), **SIAM UQ** (at Atlanta), **Ohio State** (analysis, online), **Texas A& M** (colloquium), **Drexel** (colloquium, online), **Northeastern** (colloquium, online), **UNC** (colloquium)
- **2021:** **SUNY Buffalo** (colloquium, online), **Rutgers** (special seminar, online), **Georgia Tech** (applied math, online), **UChicago** (CAM seminar), **UIUC** (PDE seminar), **Northwestern** (applied math), **CU Boulder** (waves seminar, online), **Texas A&M** (math physics), **Yale** (applied math), **MIT** (aerospace computational design seminar), **Workshop on Perturbation of Spectral Bands and Gaps** (TU Dortmund, online), **Hebrew U of Jerusalem** (analysis, online), **Tel Aviv University** (applied math, online), **SIAM MS** (online), **UC San Diego** (applied math, online), **Southern Methodist** (applied math colloquium, online), **Minnesota** (IMA Data Science, online)
- **2020:** **Maryland** (CSCAMM seminar, online), **Caltech** (CMX seminar), **UC Berkeley** (applied math)
- **2019:** **Flatiron Institute** (CCM seminar), **RPI** (colloquium), **NJIT** (waves seminar), **Tel Aviv** (applied math), **Bar Ilan** (applied math), **IMACS11** (at UGA), **Technion** (applied math), **Weizmann Institute** (applied math), **SIAM CSE** (at Spokane), **Columbia** (applied math colloquium), **Stanford** (applied math), **UC Merced** (applied math), **UC Irvine** (applied math), **CU Boulder** (waves seminar), **Yale** (applied math)
- **2018:** **SIAM NWCS** (at Orange County), **Israel Mathematical Union Annual Meeting** (at the Technion),

Contributed and other Talks

- **2025:** **NSF Comp Math Meeting** (Utah)
- **2023:** **87th Midwest PDE Seminar** (Notre Dame)
- **2022:** **Mid-Atlantic Numerical Analysis Day** (Temple), **Sayas Numerics Day** (U of Maryland Baltimore County)
- **2021:** **SIAM Annual Meeting** (online), **SIAM CSE** (online)
- **2020:** **Symposium on Machine Learning and Dynamical Systems** (Fields Institute, online), **Dynamics Days Digital** (online), **One World Waves** (online), **Dynamics Days** (Hartford)
- **2019:** **Brown-BU-UMass Dynamics and PDEs workshop** (Brown), **Young Researchers Workshop** (UMD), **OASIS 7** (Tel Aviv),
- **2017:** **Israel Physical Society Annual Meeting** (Technion), **Frontiers in Optics - OSA 101st Annual Meetings** (Washington DC)

STUDENTS SUPERVISED

- Hsin-I Hsieh (NJIT PhD), “A computer assisted approach to Floquet Hamiltonians”, (with J Jaquette)

- Sameh N. Hameedi (Columbia Applied Math M.Sc. '21, currently Ph.D. student at Oxford University). [2023 MMS paper](#). 2020–ongoing (with MI Weinstein, G-Q Chen).
- Remy Kassem (Columbia Ph.D. student), “Dispersive dynamics of the SSH model”, [2026 JMAA paper](#). 2022–ongoing (with MI Weinstein).
- Ruoxi Li (Columbia Applied Math '22). “Geometric Measure Theory” spring 2022.
- Jerry Qu (Columbia Applied Math '23). “Reproducing kernel Hilbert spaces and kernel PCA,” summer 2021 (with MI Weinstein).
- Ho Jia Xu Dion (Yale-NUS '21, currently Ph.D. student at Columbia University). “Solitary waves interactions with highly non-integrable nonlinearities,” 2019 (with W Schlag).

SERVICE AND ORGANIZATION

- **Workshop organizer** “Mathematics of Condensed Matter Physics” at ETH, Zurich (with GM Graf, J Shapiro, and MI Weinstein) 07/23
- **Referee:** Adv Math, Mem Eur Math Soc, SIAM J Math Anal, SIAM J Appl Math, SIAM J Sci Comp, SIAM J Num Anal, Comm Math Phys, SIAM/ASA J Uncert Quant, Pure Appl Anal, Bull London Math Soc, Forum Math Sigma, PRL, PRA, PRE, Phys Rev Res, J Math Phys, Ann Henry Poincare, Wave Motion, Int J Uncer Quant, Foundations of Data Science, Data-Centered Eng., Comput Stats Data Anal, JOSA B, J Math Imag Vision.
- **Grant Panel:** Binational Science Foundation (BSF)
- **Doctoral Committee**
 - Wen Ding, Columbia University, 08/22
 - Huaiyu Li, Columbia University, 08/23
 - Zirui Xu, Columbia University, 09/24
 - Edith Zhang, Columbia University, 04/25
 - Remy Kassem, Columbia University, 06/26
- **Ph.D. Proposal Committee**
 - Michael Storm, NJIT, 04/26
- **Service,** NJIT, Chair of the DMS Colloquium 2025-2026
- **Service,** Technion Faculty of Mathematics:
 - Mathematics Entry and Classification Exam 2023–2024
- **Service,** Columbia’s Department of Applied Physics and Applied Mathematics:
 - **Seminar organizer** of the APAM Friday Research Conference spring 2020, 2021, 2022
 - **Secretary of the Faculty** fall 2019 – fall 2020, spring 2022
 - **Qualifying Exams** Spring 2020, 2021, 2022, 2023
- **Mini-Course** on dynamical systems at “Columbia Summer Undergraduate Research Experiences in Mathematical Modeling” summer 2021
- **Mini-symposium and special sessions organized:**
 - “Mathematical Models in Quantum Materials” for SIAM NNP Sectional Meeting, Rutgers, Piscataway NJ (with D. Massatt) 10/26

- “Uncertainty Propagation in Dispersive and Transport dominated PDEs” for SIAM UQ26, Minneapolis MN (with L. Wang) 03/26
- “Spectra and Dynamics of Complex Materials” for SIAM/CAIMS Annual Meeting, Montreal (with E. Hiltunen) 07/25
- “Measure Transport - Algorithms and Analysis” for SIAM CSE25, Fort Worth, TX (with C. Moosmueller) 03/25
- “Computational Measure Transport” for SIAM UQ24, Trieste, Italy (with R. Baptista, A. Hsu, & B. Pandey) 02/24
- “Optimal transport in uncertainty quantification and learning” for SIAM UQ22, Atlanta, GA (with C. Moosmueller) 04/22
- “Machine Learning for Scientific Discovery” for SIAM Annual Meeting, online (with O. Lindenbaum) 07/21
- “Recent Advances in Computational Probability” for SIAM CSE21, online (with B. Hosseini) 03/21
- “Theory of Optical Waves in Novel Media” for Metamaterials 2020, online (with M.I. Weinstein) 09/20
- **Outreach talk** for undergraduate students at the Technion, “Are all functions (approximately) polynomials?” 11/23
- **Poster reviewer** at an undergraduate poster research session, JMM 2026.
- **Conference referee** for Metamaterials 2020 and Metamaterials 2021
- **Tutoring** for undergraduate students with physical disabilities and for supporting foreign students. Tel Aviv University 2016-2017

TEACHING EXPERIENCE

New Jersey Institute of Technology

- Introduction to Mathematical Analysis (undergraduate and graduate, MATH480/545) fall 2026
- Linear Algebra (undergraduate, MATH337), fall 2025
- Complex Variables I (graduate, MATH656) spring 2025, 2026

Technion

- Introduction to Applied Mathematics (undergraduate, 00104192) spring 2024
- Asymptotic Analysis (graduate, 00198000) spring 2024

Columbia University

- Multivariate Calculus for Engineering and Applied Sciences (APAM2000E) fall 2019, 2020, 2021
- Principles of Applied Mathematics (APMA4001E) spring 2020
- Applied Mathematics III: Dynamical Systems (APMA4101E) spring 2021, 2022, 2023

Tel Aviv University

- Numerical Analysis for Engineering spring 2018

Tel Aviv University - Teaching Assistant

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| • Numerical Analysis | fall 2017 |
| • Ordinary Differential Equations | spring 2017 |
| • Calculus I | fall 2017 |
| • Ordinary Differential Equations for Engineering | spring 2016 |